

Q1.

A car travels around a roundabout at a constant speed. The surface of the roundabout is horizontal.

The car has mass 990 kg and the path of the car is a circular arc of radius 48 metres.

A simple model assumes that the car is a particle and the only horizontal force acting on it as it travels around the roundabout is friction.

On a dry day typical values of friction, F , between the surface of the roundabout and the tyres of the car are

$$7300 \text{ N} \leq F \leq 9200 \text{ N}$$

- (a) Using this model calculate a safe speed limit, **in miles per hour**, for the car as it travels around the roundabout.

Explain your reasoning fully.

Note that there are 1600 metres in one mile.

(4)

- (b) Gary assumes that on a wet day typical values for friction, F , are

$$5400 \text{ N} \leq F \leq 10\,000 \text{ N}$$

Comment on the validity of Gary's revised assumption.

(2)

(Total 6 marks)

Q2.

A child, of mass 40 kg, moves at constant speed of 5.0 m s^{-1} on a fairground ride.

The path of the child is a circle of radius 4.0 metres.

Find the magnitude of the resultant force acting on the child.

Circle your answer.

6.3 N

50 N

130 N

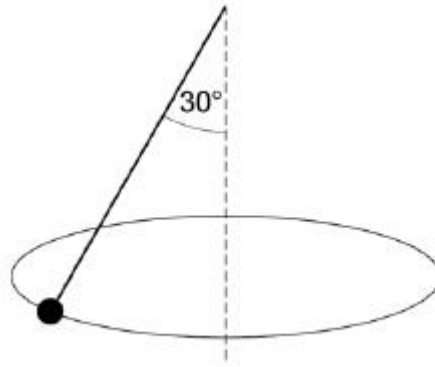
250 N

(Total 1 mark)

Q3.

In this question use $g = 9.8 \text{ m s}^{-2}$.

A conical pendulum consists of a string of length 60 cm and a particle of mass 400 g. The string is at an angle of 30° to the vertical, as shown in the diagram.

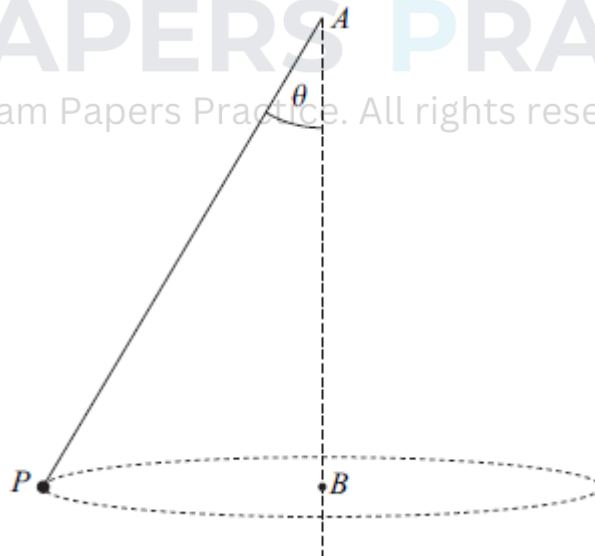


- (a) Show that the tension in the string is 4.5 newtons. (2)
- (b) Find the angular speed of the particle. (3)
- (c) State **two** assumptions that you have made about the string. (1)
- (Total 6 marks)

Q4.

A light inextensible string has one end attached to a particle, P , of mass 2 kg. The other end of the string is attached to the fixed point A . The point A is vertically above the point B . The particle moves at a constant speed in a horizontal circle of radius 0.8 m and centre B . The tension in the string is 34 N.

The string is inclined at an angle θ to the vertical, as shown in the diagram.



- (a) Find the angle θ . (3)
- (b) Find the speed of the particle. (3)

- (c) Find the time taken for the particle to make one complete revolution.

(2)

(Total 8 marks)

Q5.

Tom is travelling on a train which is moving at a constant speed of 15 m s^{-1} on a horizontal track. Tom has placed his mobile phone on a rough horizontal table. The coefficient of friction between the phone and the table is 0.2.

The train moves round a bend of constant radius. The phone does not slide as the train travels round the bend.

Model the phone as a particle moving round part of a circle, with centre O and radius r metres.

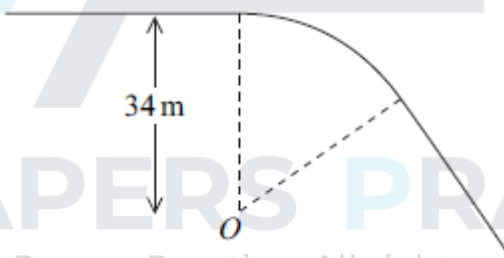
Find the least possible value of r .

(Total 4 marks)

Q6.

A parcel is placed on a flat rough horizontal surface in a van. The van is travelling along a horizontal road. It travels around a bend of radius 34 m at a constant speed. The coefficient of friction between the parcel and the horizontal surface in the van is 0.85.

Model the parcel as a particle travelling around part of a circle of radius 34 m and centre O , as shown in the diagram.



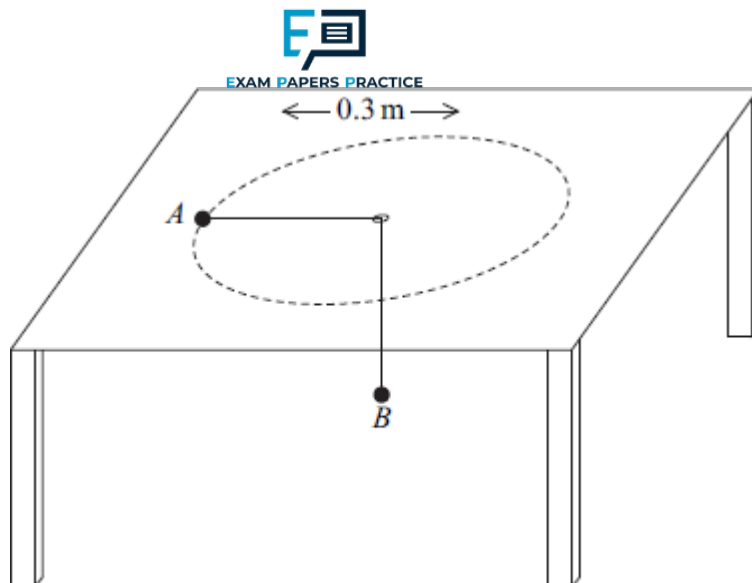
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Find the greatest speed at which the van can travel around the bend without causing the parcel to slide.

(Total 6 marks)

Q7.

Two particles, A and B , are connected by a light inextensible string which passes through a hole in a smooth horizontal table. The edges of the hole are also smooth. Particle A , of mass 1.4 kg, moves, on the table, with constant speed in a circle of radius 0.3 m around the hole. Particle B , of mass 2.1 kg, hangs in equilibrium under the table, as shown in the diagram.



- (a) Find the angular speed of particle A . (4)
 - (b) Find the speed of particle A . (2)
 - (c) Find the time taken for particle A to complete one full circle around the hole. (2)
- (Total 8 marks)

Q8.

- (a) A shiny coin is on a rough horizontal turntable at a distance 0.8 m from its centre. The turntable rotates at a constant angular speed. The coefficient of friction between the shiny coin and the turntable is 0.3 .

Find the maximum angular speed, in radians per second, at which the turntable can rotate if the shiny coin is not going to slide.

(4)

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- (b) The turntable is stopped and the shiny coin is removed. An old coin is placed on the turntable at a distance 0.15 m from its centre. The turntable is made to rotate at a constant angular speed of 45 revolutions per minute.

- (i) Find the angular speed of the turntable in radians per second.

(2)

- (ii) The old coin remains in the same position on the turntable.

Find the least value of the coefficient of friction between the old coin and the turntable needed to prevent the old coin from sliding.

(4)

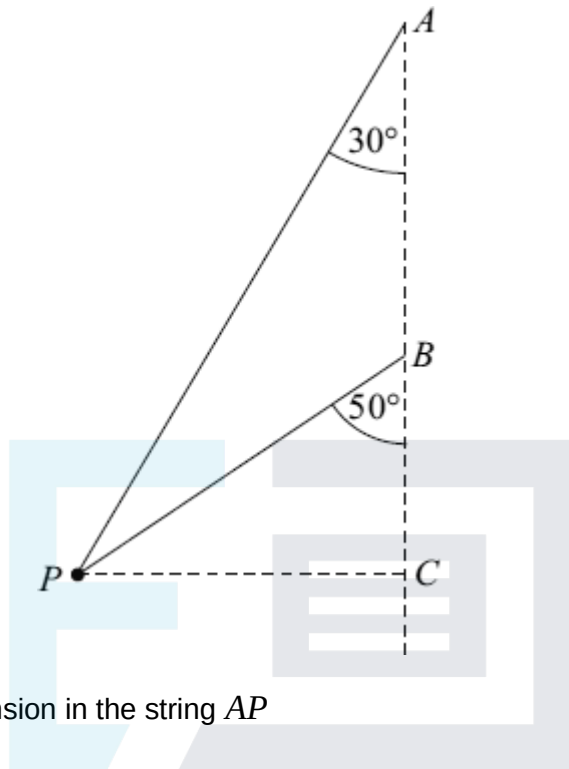
(Total 10 marks)

Q9.

Two light inextensible strings each have one end attached to a particle, P , of mass 4 kg . The other ends of the strings are attached to the fixed points A and B . The point A is vertically above the point B .

The particle moves at a constant speed in a horizontal circle. The centre, C , of this circle is directly below the point B . The two strings are inclined at 30° and 50° to the vertical, as shown in the diagram. Both strings are taut.

As the particle moves in the horizontal circle, the tension in the string BP is 20 N.



- (a) Find the tension in the string AP

(4)

- (b) The speed of the particle is 5 m s^{-1} .

Find the length of CP , the radius of the horizontal circle.

(4)

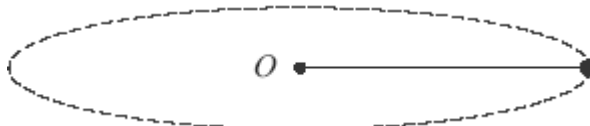
(Total 8 marks)

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Q10.

A particle, of mass 8 kg, is attached to one end of a length of elastic string. The particle is placed on a smooth horizontal surface. The other end of the elastic string is attached to a point O fixed on the horizontal surface.

The elastic string has natural length 1.2 m and modulus of elasticity 192 N.



The particle is set in motion on the horizontal surface so that it moves in a circle, centre O , with constant speed 3 m s^{-1} .

Find the radius of the circle.

(Total 8 marks)

Q11.

A car of mass 1200 kg travels round a roundabout on a horizontal, circular path at a constant speed of 14 m s^{-1} . The radius of the circle is 50 metres. Assume that there is no resistance to the motion of the car and that the car can be modelled as a particle.

- (a) A friction force, directed towards the centre of the roundabout, acts on the car as it moves. Show that the magnitude of this friction force is 4704 N.

(4)

- (b) The coefficient of friction between the car and the road is μ . Show that $\mu \geq 0.4$.

(3)

(Total 7 marks)

Q12.

A motorcyclist rides her motorcycle at a constant speed on a horizontal circular path around a roundabout.

- (a) The radius of the circular path of the motorcycle is 15 metres.

- (i) Calculate the magnitude of the acceleration of the motorcycle when it is moving at 7.5 m s^{-1} .

(2)

- (ii) The total mass of the motorcycle and rider is 400 kg. To avoid skidding taking place, the magnitude of the frictional force between the motorcycle and the road must not exceed 2940 N. Find the greatest speed, $V \text{ m s}^{-1}$, at which the motorcyclist can ride her motorcycle around this roundabout without skidding.

(3)

- (b) Without referring to air resistance, state one modelling assumption you have made in answering part (a). Explain why your assumption is reasonable.

(2)

- (c) In wet weather the maximum frictional force between the motorcycle and the road is reduced. The motorcyclist wishes to ride her motorcycle at the speed $V \text{ m s}^{-1}$, as found in part (a)(ii), around the roundabout. State whether she should increase or decrease the radius of her circular path to avoid skidding, giving a reason for your answer.

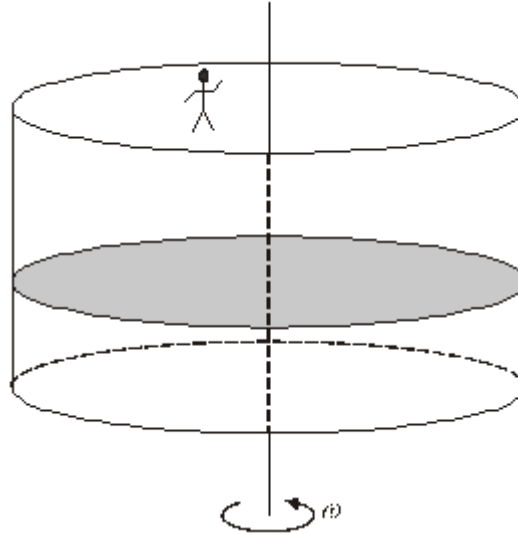
(2)

(Total 9 marks)

Q13.

A ride in an amusement park consists of a hollow cylindrical cage of radius 3 m, with a horizontal floor and vertical sides.

Martin, who has a mass of 40 kg, stands at the edge of the floor of the cage, against the wall. The cage begins to rotate about its vertical axis. When the cage is rotating at a constant angular speed, ω radians per second, the cage floor is lowered. Martin remains in the position shown in the diagram due to the vertical frictional force between him and the wall.

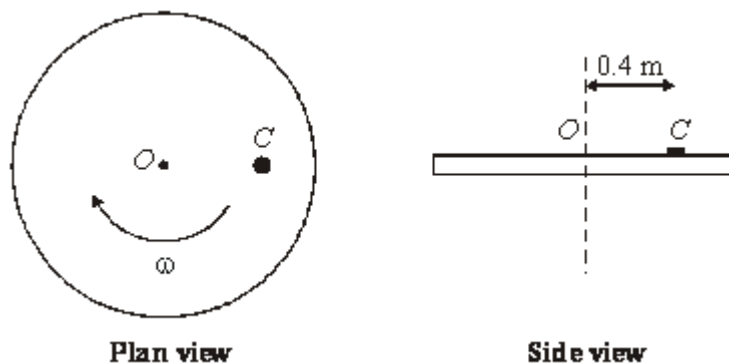


- (a) Draw a diagram showing the forces acting on Martin after the floor has been lowered. (1)
- (b) (i) State the magnitude of the frictional force. (1)
- (ii) The magnitude of the normal reaction between Martin and the wall is 784 N. Find the least possible value of the coefficient of friction between Martin and the wall. (2)
- (iii) Show that $\omega = 2.56$, correct to three significant figures. (3)
- (c) State **one** modelling assumption used in this question. (1)
- (Total 8 marks)

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Q14.

A small coin, C , is placed on a rough horizontal turntable, as shown in the diagram.



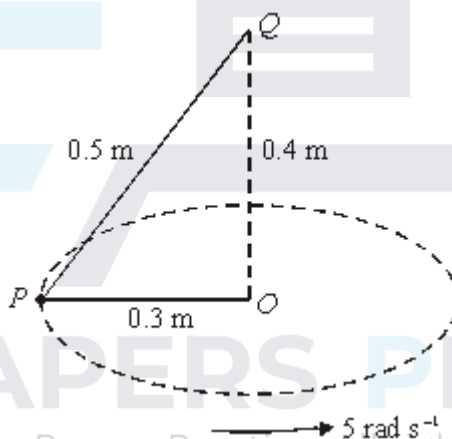
The mass of the coin is 0.01 kg and it is placed at a distance of 0.4 m from O , the centre of the turntable.

The turntable rotates about a vertical axis through O , with a constant angular speed of ω radians per second. During this motion the coin does **not** slip.

- (a) Draw a diagram to show all the forces acting on the coin. (1)
 - (b) Find, in terms of ω , an expression for the frictional force acting on the coin. (2)
 - (c) The coefficient of friction between the coin and the turntable is 0.8. The coin is on the point of slipping. Show that the value of ω is approximately 4.43. (4)
- (Total 7 marks)**

Q15.

A particle of mass 0.4 kg is attached at the point P to two light strings, QP and OP . The points O and Q are fixed with Q at a distance of 0.4 m vertically above O . The string QP is inextensible and of length 0.5 m. The string OP is elastic and of natural length 0.2 m and stiffness $k \text{ N m}^{-1}$. The particle moves in a horizontal circle, centre O and radius 0.3 m, at a constant angular speed of 5 rad s^{-1} .



- (a) Draw a diagram showing the forces acting on the particle. (1)
 - (b) Show that the tension in the string QP is 4.9 N. (3)
 - (c) Write down, in terms of k , the tension in the string OP . (1)
 - (d) Show that $k = 0.6$. (5)
 - (e) Find the elastic potential energy stored in the string OP . (2)
- (Total 12 marks)**

Q16.

The orbit of the moon around the earth may be modelled as circular. The time taken for the moon to make one complete orbit is approximately 27.32 days.

- (a) Show that the angular speed of the moon is approximately 2.66×10^{-6} radians per second.

(3)

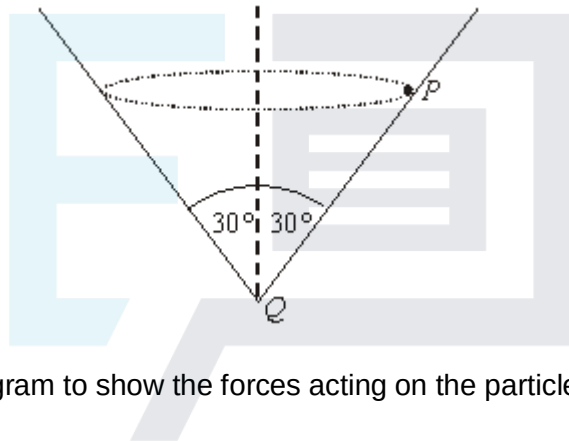
- (b) Assuming that the radius of the circular orbit is approximately 3.844×10^8 metres, find the speed of the moon, relative to the earth, in metres per second.

(2)

(Total 5 marks)

Q17.

A particle P , of mass m , moves in a horizontal circular path on the inside of a smooth cone with vertex Q . The angle between PQ and the vertical is 30° . The diagram shows the cone and the path of the particle.



- (a) Draw a diagram to show the forces acting on the particle.

(1)

- (b) Show that the reaction force on the particle has magnitude $2mg$.

(2)

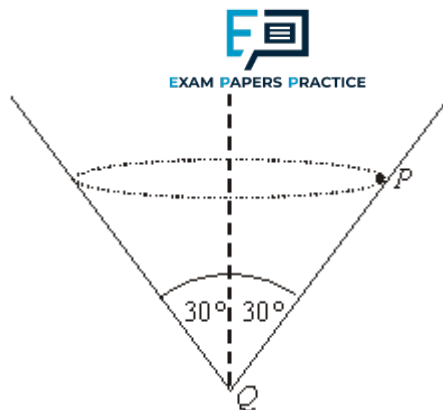
- (c) The speed of the particle is v . Find the radius, r , of the circular path of P in terms of v and g .

(4)

(Total 7 marks)

Q18.

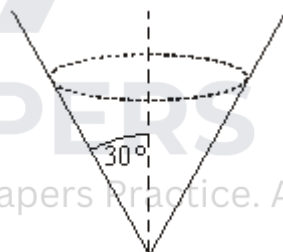
A particle P , of mass m , moves in a horizontal circular path on the inside of a smooth cone with vertex Q . The angle between PQ and the vertical is 30° . The diagram shows the cone and the path of the particle.



- (a) Draw a diagram to show the forces acting on the particle. (1)
- (b) Show that the reaction force on the particle has magnitude $2mg$. (2)
- (c) The speed of the particle is v . Find the radius, r , of the circular path of P in terms of v and g . (4)
- (d) Explain what would happen to the radius if the speed were halved. (2)
- (Total 9 marks)

Q19.

A particle, of mass 3 kg , describes a horizontal circular path on the inside surface of a smooth cone, as shown in the diagram.



The radius of the circle is 0.5 metres and the semi-vertical angle of the cone is 30° . The particle moves at a constant speed.

- (a) (i) Show that the magnitude of the normal reaction force on the particle is 58.8 N . (3)
- (ii) Find the speed of the particle. (4)
- (b) The particle moves on the same cone, but in a horizontal circle of greater radius than before.
- (i) What happens to the magnitude of the normal reaction force? (1)
- (ii) What happens to the speed of the particle? Explain your answer.



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